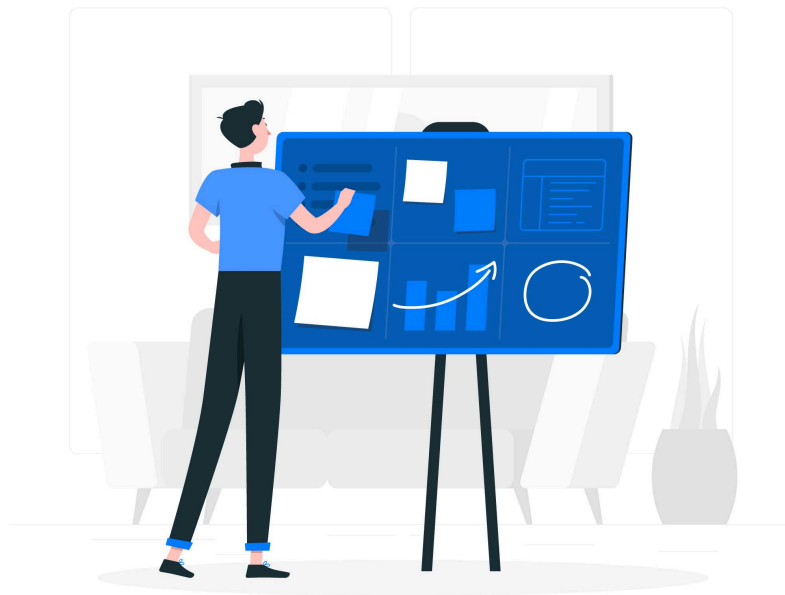


# Malware Analysis Project – Behavioral and Memory Forensics

## 1. General Description of the Task:

This project involves conducting a comprehensive behavioral and memory forensic analysis of a real-world malware sample in a controlled, virtualized lab environment. The objective is to observe, capture, and interpret the malware's runtime behavior, network communication patterns, memory artifacts, and persistence mechanisms.

Participants will execute the malware within an isolated Windows virtual machine, while capturing both network traffic and a full memory dump during the infection process. The analysis will be carried out on a separate Linux-based forensic system using tools like Wireshark, FTK Imager, and Volatility.



## 2. Objectives:

- Perform a full behavioral analysis of a **ZeroAccess Trojan** attached.
- Safely execute the malware inside a controlled and isolated **Windows virtual machine**.
- Capture and analyze **network traffic** and **memory dumps**.
- Identify Indicators of Compromise (IOCs), malware behavior, and persistence mechanisms.
- Map observed actions to the **MITRE ATT&CK framework**.
- (Optional) Create a **YARA rule** to detect the malware.



### 3. Tools to be Used:

- **Inside Windows VM (Isolated):**
  - ☒ Wireshark – For network traffic capture.
  - ☒ FTK Imager – For memory acquisition.
- **On Kali Linux Analysis Machine:**
  - ☒ Volatility – For memory analysis.
  - ☒ Any additional tools.



## 4. Requirements:

### 4.1. Virtual Environment Setup

#### Requirement Details/Steps:

- Set up a clean **Windows 7 or 10 VM**.
- Disable Windows Defender, auto-updates, and internet access.
- Copy and **extract the malware sample** using the password infected.
- Guide: <https://www.wikihow.com/Install-Windows-7-on-a-VMware-Workstation>

#### Deliverable of this Requirement:

- A brief setup summary in the final report confirming isolation and prep steps.

### 4.2. Network Traffic Monitoring

#### Requirement Details/Steps:

- Launch **Wireshark** before executing the malware.
- Begin traffic capture and let it run in the background.
- Execute the malware and let it operate for **60 seconds**.
- Stop and save the capture as malware\_analysis.pcap.

#### Deliverable of this Requirement:

- Wireshark .pcap file containing all network activity during execution.

### 4.3. Memory Dump Capture

#### Requirement Details/Steps:

- Use FTK Imager to perform a **full memory dump**.
- Save the output as `memory.raw`.

#### Deliverable of this Requirement:

- Full memory dump file `memory.raw` to be used for further analysis.

### 4.4. Forensic Analysis

#### Requirement Details/Steps:

- Transfer both `memory.raw` and `malware_analysis.pcap` to **Kali Linux**.
- Analyze network activity (e.g., external IPs, DNS requests, C2 behavior).
- Use **Volatility** for memory forensics:
  - Investigate processes, DLL injection, registry modifications.
  - Look for file drop activity and persistence techniques.
- Document findings: timeline of events, affected processes, and file paths.

#### Deliverable of this Requirement:

- Documented behavior analysis including:
  - Network Indicators
  - Memory Artifacts
  - Persistence Mechanisms
  - IOCs



## 5. Final Deliverables:

- ☒ `malware_analysis.pcap` – Captured network traffic
- ☒ `memory.raw` – Full memory image
- ☒ `report.pdf` – Detailed report including:
  - Execution summary
  - Behavioral observations
  - Network and memory analysis
  - IOCs



